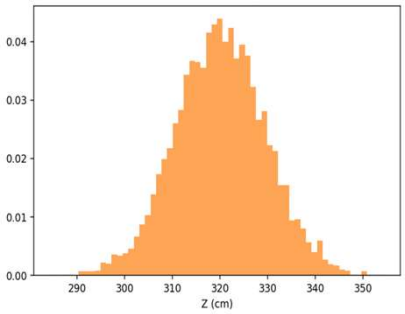
Agenda

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Last time: Why normal?
  - z-score and proportions
  - Sums of
    - independent but
    - not identical
    - normal distribution
  - The sum is normal
  - Central Limit Theorem



2

Sums of independent normal distribution

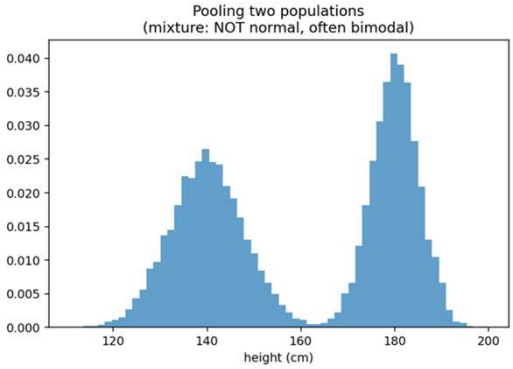
- $Y = X_1 + X_2 + \cdots X_n$ where each $X_i \sim N(\mu_i, \sigma_i^2)$ and *independent*
  - Answer: $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$
- Per theorem,
  - What happens if measure heights X_1 of 100 selected national basketball players (normal distribution)
  - and, say, 100 kho-kho players X_2 (also normal distribution)
  - Assume independence
  - Put them all in this room
  - Distribution?

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

3

Sums of independent normal distribution

- $Y = X_1 + X_2 + \cdots X_n$ where each $X_i \sim N(\mu_i, \sigma_i^2)$ and independent
- Answer: $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$
- Per theorem,
  - what happens if measure heights X_1 of 100 selected national basketball players
  - and, say, 100 kho-kho players X_2?
  - Put them all in this room
  - Distribution?



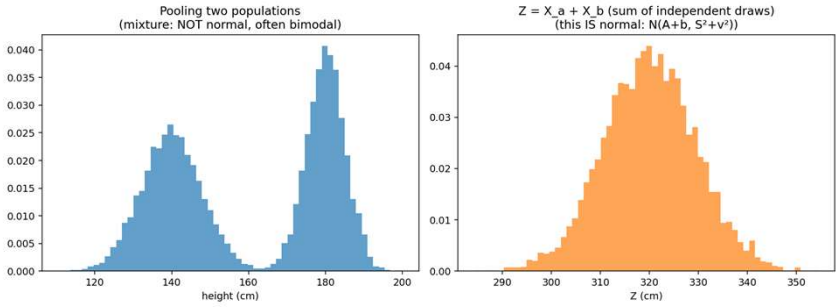
5

Sums of independent normal distribution

- $Y = X_1 + X_2 + \cdots X_n$ where each $X_i \sim N(\mu_i, \sigma_i^2)$ and independent
- Answer: $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$
- Per theorem, what happens if measure heights X_1 of national basketball players and, say, kho-kho players X_2?

The result is not about concatenation, or mixture

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

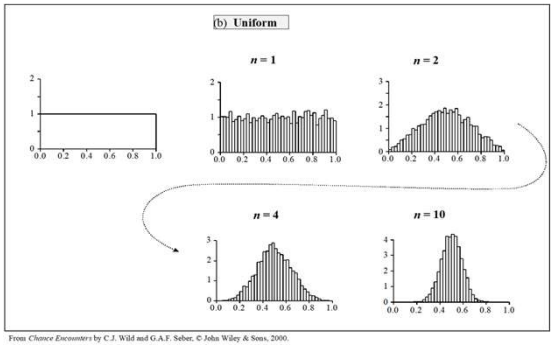


6

Agenda

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Last time: Why normal?
  - z-score, proportions
  - Sums of independent but not identical normal distribution
  - Sums of independent identical distributions, not necessarily normal (Central Limit Theorem)



From *Chance Encounters* by C.J. Wild and G.A.F. Seber, © John Wiley & Sons, 2000.

7

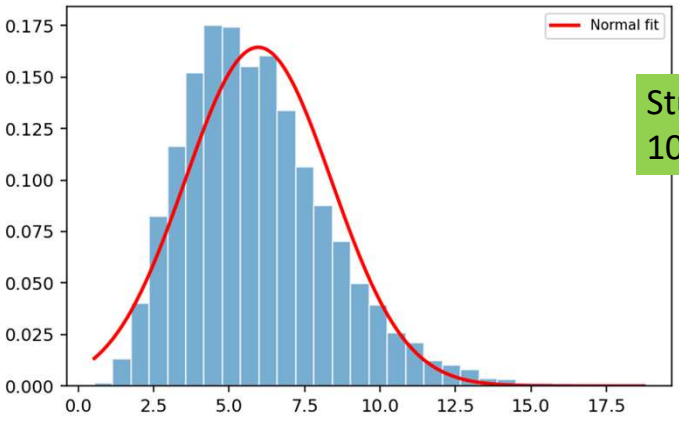
Why normal?

- For independent identical random variables, and $Y = X_1 + X_2 + \cdots X_n$
- Central limit theorem
  - For large n, the distribution of Y(n) is approximately normal with mean and variance as earlier seen as if the distributions were normal
- How large is n?

The statement of the theorem is not precise

8

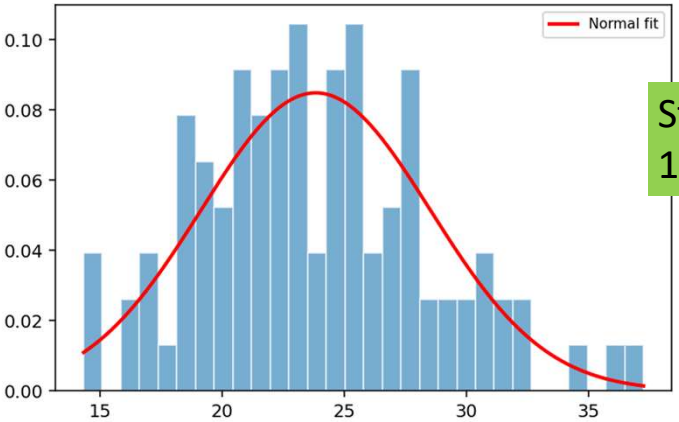
Which distribution is “more” normal?



Student A: n = 6, 10000 experiments

9

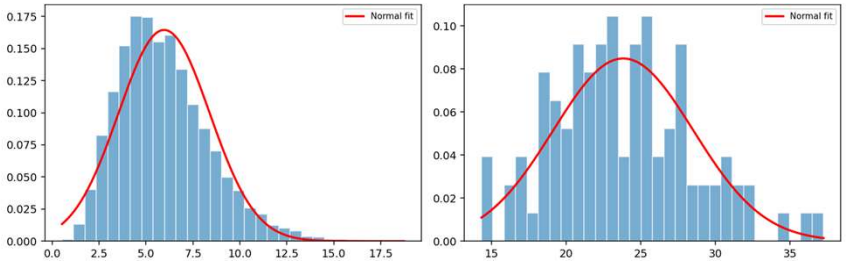
Which distribution is “more” normal?



Student B n=24, 100 experiments

10

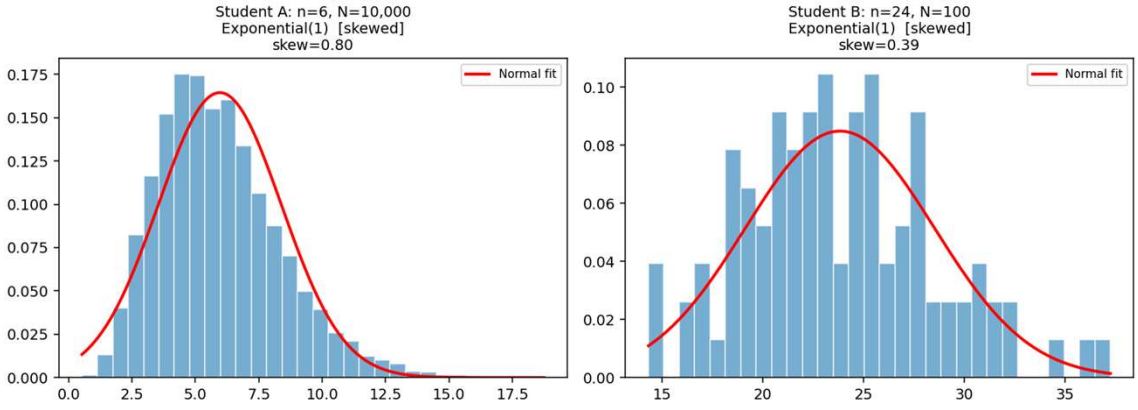
Which distribution is “more” normal?



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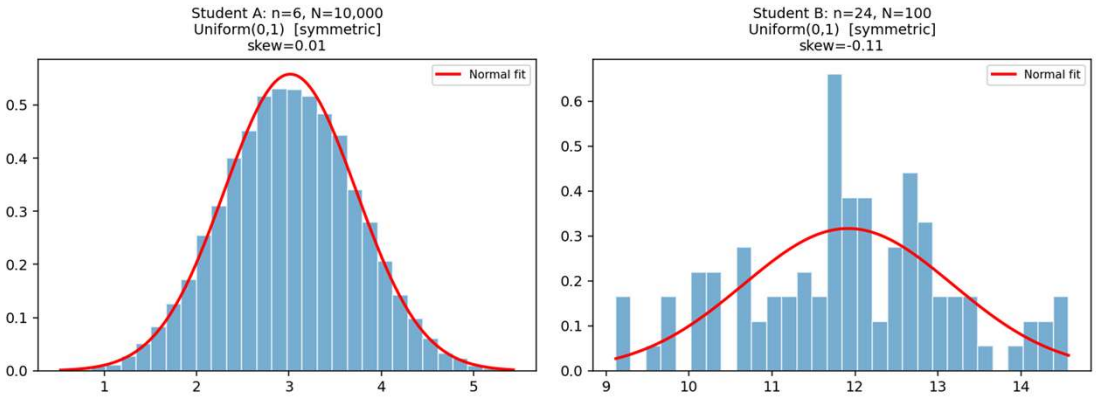
11

Which distribution is “more” normal?



13

But …



14

Why normal?

- For independent identical random variables, and $Y = X_1 + X_2 + \cdots X_n$
- Central limit theorem
  - For large (but finite) n, the distribution of Y is approximately normal with mean and variance as earlier seen as if the distributions were normal
  - Finite n, sample mean $\overline{x} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
  - Notice that it helps to know the mean and variance of typical distributions corresponding to X_i

Takeaway message

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Normal and Sums of Bernoulli

- Let X= $X_1 + X_2 + \cdots X_n$
  - Each $X_i \sim \text{Bern}(p)$
- Clearly $X \sim \text{Bin}(n, p)$
- Exact distribution
- Central limit theorem approximation:
  - For large n, $X \sim \mathcal{N}(np, np(1-p))$
  - $\frac{Y-np}{\sqrt{np(1-p)}}$ is standard normal

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

$$E(X) = np$$

$$\text{Var}(X) = np(1-p)$$

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$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

Example: Normal and Binomial



- Your CR is organizing “garba” in new-new-sac (100 size)
  - From past experience, students who register turn up with a probability of 0.7
  - Your CR wants to “overbook” the event maximizing cover fee
  - How many registrations can he accept if the probability of more than 100 showing up is 5%?
- X number of students: $\text{Bin}(n, 0.7)$
  - What is the largest n for “good case” $\longrightarrow P(X \geq 101) \leq 0.05$)

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$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

Example: Normal and Binomial

- Your CR is organizing "garba" in new-new-sac (100 size)
  - From past experience, students who register turn up with a probability of 0.7
  - Your CR wants to "overbook" the event maximizing cover fee
- X number of students: Bin$(n, 0.7)$
- What is the largest n for "good case" $\longrightarrow$ $P(X \geq 101) \leq 0.05)$
- Binomial: For each n,
  - Evaluate $P(X \geq 101) = \sum_{k=101}^{n} \binom{n}{k} (0.7)^k (0.3)^{n-k}$
  - For example, with n = 131, $P(X \geq 101) = \sum_{k=101}^{131} \binom{131}{k} (0.7)^k (0.3)^{131-k} \approx 0.0441$

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Example: Normal and Binomial

$$E(X) = np$$

$$\mathrm{Var}(X) = np(1-p)$$

- How many registrations can he accept if the probability of more than 100 showing up is 5%?
  - $P(X \geq 101) \leq 0.05)$
- Normal approximation $X \approx \mathcal{N}(0.7n, 0.21n)$
  - Continuity correction
  - Map to z-score
  - Inverse probability and one-sided tail
  - Solve quadratic equation
  - Complicated binomial sum has same answer as quadratic equation

$$P(X \geq 101) \approx P(Y > 100.5),$$

$$P\left(Z > \frac{100.5 - 0.7n}{\sqrt{0.21n}}\right) \leq 0.05.$$

$$P(Z > 1.645) = 0.05,$$

$$\frac{100.5 - 0.7n}{\sqrt{0.21n}} \geq 1.645.$$

$$n \approx 131.24.$$

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